Highlights

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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- Closed-loop knowledge-guided swarm learning framework for high-dimensional feature selection.
- MI initialization with adaptive re-initialization restores diversity and prevents search entrapment.
- Knowledge-guided elite refinement accumulates feature-importance signals with exponential decay.
- Two-phase weight scheduling temporally decouples feature compression from accuracy refinement.
- KDS-EL achieves 95.74% accuracy with 87.8 selected features across 20 benchmark datasets.

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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ABSTRACT

Feature selection in high-dimensional data remains a central challenge for evolutionary computation. Particle swarm optimization suffers from three structural deficiencies: random initialization provides insufficient guidance in sparse high-dimensional spaces, uniform feature perturbation fails to exploit cross-iteration feature-importance signals, and static objective preferences cannot adapt to different evolutionary phases. To address these limitations, we propose KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a knowledge-guided evolutionary framework with a closed-loop feature-importance learning mechanism. First, to overcome inefficient initialization, we propose mutual-information-guided initialization with adaptive re-initialization, injecting prior knowledge and recovering population diversity when stagnation occurs. Second, to exploit discarded cross-iteration feature-importance signals, we propose a knowledge-guided elite refinement strategy that accumulates feature-importance knowledge and steers elite solutions toward historically effective features. Third, to align objective weights with search phases, we propose two-phase weight scheduling that emphasizes early dimensionality reduction and late-stage accuracy refinement. On 20 high-dimensional benchmarks, KDS-EL achieves the best classification performance among all competitors, with a mean accuracy of 0.9574 and only 87.8 selected features, outperforming the strongest single- and multi-objective competitors by +3.97 and +4.42 percentage points; Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm statistically significant improvements over all 13 competitors.

1. Introduction

High-dimensional data have emerged as a fundamental challenge in classification problems across diverse domains, including genomics, text mining, and computer vision. These datasets typically contain hundreds or even tens of thousands of features, while only a limited number of labeled samples are available, resulting in the well-known $p \gg n$ scenario. Under such conditions, conventional learning algorithms often suffer from severe overfitting, increased computational complexity, and reduced interpretability due to the curse of dimensionality [**?** **?**]. Feature selection (FS) has therefore become an essential preprocessing technique that aims to identify a compact and informative subset of features while preserving classification performance. However, selecting an optimal feature subset remains highly challenging because the search space grows exponentially with the number of features D, and the underlying subset-selection problem is NP-hard [**?**]. Moreover, individual feature relevance measures are insufficient to characterize complex feature interactions and dependencies, which are often critical for achieving robust classification performance.

FS methods are generally categorized into filter, wrapper, and embedded approaches [**?** **?**]. Filter methods rank features according to statistical criteria and offer high computational efficiency, but they usually evaluate features independently of the classifier and may overlook feature interactions relevant to classification. Embedded methods incorporate feature selection into the learning process, providing a compromise between selection efficiency and predictive performance; however, their effectiveness is often dependent on the characteristics of specific learning models. Wrapper methods directly optimize feature subsets according to classifier performance and can achieve competitive classification results, but the repeated model evaluations make them computationally expensive, especially in high-dimensional search spaces. To address the combinatorial complexity of wrapper-based FS, evolutionary computation has been extensively investigated as a promising optimization paradigm [**?** **?** **?** **?**]. By maintaining a population of candidate solutions, evolutionary algorithms can perform global exploration of feature subsets without requiring gradient information. Among them, particle swarm optimization (PSO), genetic algorithms, and differential evolution have demonstrated promising results on high-dimensional FS tasks.

PSO has become one of the most widely adopted evolutionary algorithms for FS due to its simple structure, efficient information-sharing mechanism, and fast convergence. Nevertheless, conventional PSO-based FS methods still encounter three structural limitations when dealing with sparse and high-dimensional search spaces. First, random initialization distributes particles without prior guidance throughout the binary search space, whereas effective feature

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subsets usually occupy only a small region with limited active dimensions. Consequently, a considerable number of fitness evaluations may be wasted on redundant and suboptimal solutions before the swarm identifies promising search regions. Second, conventional uniform feature perturbation strategies treat all dimensions equally and neglect the feature-level improvement information accumulated during previous evaluations, preventing the search from exploiting historically informative features. Third, existing PSO-based FS methods typically employ fixed objective weights to balance classification error and feature subset size during the entire optimization process. However, the desired trade-off between feature reduction and accuracy refinement varies across search stages: aggressive sparsity enhancement is preferred in early exploration, whereas accurate decision boundary refinement becomes more important after a compact subset has been obtained.

Recent evolutionary FS studies have attempted to improve search performance through enhanced population initialization, adaptive evolutionary operators, and refined fitness evaluation strategies [**? ? ? ?**]. Although these approaches have achieved promising results, they usually focus on isolated components of the optimization process and do not jointly address inefficient initialization, insufficient exploitation of accumulated feature knowledge, and inflexible objective weighting. Recently, multi-task PSO has emerged as an effective paradigm for high-dimensional FS with fixed-length binary representations. For instance, MEL [**?**] adopts cooperative subpopulations and a fixed-length representation to efficiently handle feature spaces containing tens of thousands of dimensions. Despite its effectiveness, existing multi-task PSO-based FS frameworks still leave several challenges unresolved: the initial population is generally generated without sufficient feature-level prior knowledge, useful feature-importance information discovered during evolution is not explicitly retained and exploited, and the balance between classification performance and feature sparsity remains unchanged across different search stages.

To address these limitations, this paper proposes KDS-EL (Knowledge-Guided Dual-Phase Swarm Evolutionary Learning), a knowledge-guided evolutionary framework with a closed-loop feature-importance learning mechanism that integrates prior knowledge injection, evolutionary knowledge accumulation, and adaptive objective scheduling into the multi-task PSO architecture. The proposed framework consists of three complementary mechanisms that address different aspects of high-dimensional FS optimization.

First, mutual-information-guided initialization with adaptive re-initialization incorporates prior feature-relevance knowledge into population construction, reduces early-stage search waste caused by unguided initialization, and restores population diversity when stagnation occurs. Second, the Knowledge-guided Elite Refinement (KER) strategy accumulates cross-iteration feature-importance knowledge from fitness comparisons performed during evolution, applies exponential decay to reduce the influence of outdated evidence, and guides elite solutions toward historically informative features within a bounded refinement budget. Third, Two-Phase Weight Scheduling (TPS) dynamically adjusts the dual-objective preference from feature reduction to classification refinement, matching optimization priorities with the evolving requirements of high-dimensional FS. These mechanisms operate at different levels of the evolutionary process, and their individual contributions can be systematically verified through component contribution analysis.

The main contributions of this paper are summarized as follows:

1. An MI-guided initialization scheme with adaptive re-initialization is proposed to incorporate feature-relevance knowledge into population construction and restore population diversity when evolutionary stagnation occurs.
2. A Knowledge-guided Elite Refinement (KER) strategy is developed to accumulate cross-iteration feature-importance knowledge from evolutionary feedback, reduce outdated information through exponential decay, and guide elite solutions toward historically effective features within a bounded refinement budget.
3. A Two-Phase Weight Scheduling (TPS) mechanism is introduced to replace static objective weighting with a stage-aware optimization strategy, emphasizing aggressive feature reduction in early evolution and accuracy refinement in later evolution.
4. Comprehensive experiments are conducted on 20 high-dimensional benchmark datasets containing 500 to 54,675 features against 14 fixed-length evolutionary FS methods. Statistical tests, ablation studies, and parameter sensitivity analyses are performed to evaluate the effectiveness and robustness of KDS-EL.

Extensive experiments demonstrate that KDS-EL achieves the best classification performance, obtaining an average classification accuracy of 0.9574 with only 87.8 selected features. Compared with the strongest single-objective and multi-objective competitors, KDS-EL improves accuracy by 3.97 and 4.42 percentage points, respectively, while achieving a 3.4× speedup over the MEL baseline (reducing average runtime from 175.0 s to 51.4 s). Furthermore, Wilcoxon signed-rank tests with Holm–Bonferroni correction confirm that the improvements are statistically significant over all 13 competitors.

The remainder of this paper is organized as follows. Section **??** reviews feature selection methods and evolutionary FS algorithms. Section **??** introduces the background knowledge, including PSO, mutual-information-based feature relevance estimation, and the MEL framework. Section **??** presents the proposed KDS-EL framework and describes its three mechanisms in detail. Section **??** reports the experimental configuration and results, including comparative experiments, component contribution analysis, and parameter sensitivity analysis. Finally, Section **??** concludes this paper and discusses future research directions.